Milestone M3 (Lane B): Formal Observer Validation Protocol

Project: IIT/SRIC/R/AEH/2026/104 (CMPDI) — *Dynamic Modelling and Digital Twin of Electric Cable Shovel* **Code Version:** 2.0.0-PRELIMINARY (m3_observer_suite.m & run_m3_experiments_v2.py) **Sampling Rate:** $f_s = 30\ \mathrm{Hz}$ ($T_s = 1/30\ \mathrm{s} = 33.3333\ \mathrm{ms}$) **Frozen Nominal Gains:** $K_O = \mathrm{diag}(20, 20)\ \mathrm{s}^{-1}$, Branch A RK4 substeps $M = 67$ ($\Delta t = 0.4975\ \mathrm{ms}$)

1. Mandatory Pre-Execution Data Ingestion & Integrity Gate (load_and_validate_mat_archive)

Before any observer simulation is executed, load_and_validate_mat_archive(filepath, P) enforces 7 automated failure gates:

1. **File Existence & Format Check**: Rejects missing files immediately (m3_observer_suite:FileNotFound) without falling back to synthetic trajectories.
2. **Summary-Struct Rejection Gate**: Rejects summary-only MAT files such as stage1_results.mat (m3_observer_suite:SummaryOnlyMatFile) that lack time-series vectors.
3. **Required Signal Completeness**: Verifies all 20 required signals in R.sig (q3, d4, q3d, d4d, q3dd, d4dd, tau3_meas, tau3_model, e_tau3, f4_meas, f4_model, e_f4, Fx, Fy, Qq3_pred, Qd4_pred, ref3, ref4, s, r).
4. **Timestamp Monotonicity & Uniformity**: Confirms $N_{\mathrm{raw}} = 3{,}001$ samples at uniform $f_{s,\mathrm{raw}} = 100.0\ \mathrm{Hz}$ ($t \in [0, 30]\ \mathrm{s}$, trim window $t \in [5.0, 30.0]\ \mathrm{s}$).
5. **Parameter Lock Verification**: Confirms R.params matches shovel_params.m ($M_d = 54{,}300\ \mathrm{kg}$, $I_{zz} = 287{,}900\ \mathrm{kg\cdot m^2}$, $c = -1.32\ \mathrm{m}$, $f_{v3} = 3{,}800$, $f_{v4} = 1{,}473$, $g = 9.80665$).
6. **Kinematic Station Verification**: Confirms R.sig.r == d4 - 1.32 and R.sig.s == d4 + R.opt.Station to $< 10^{-9}\ \mathrm{m}$.
7. **Spline Resampling Consistency**: Verifies that resampling from $100\ \mathrm{Hz}$ to $30\ \mathrm{Hz}$ ($N_{30} = 901$, $N_{\mathrm{eval}} = 751$ over $t \in [5, 30]\ \mathrm{s}$) preserves the inverse-dynamics equation to $\le 3.59\times 10^{-5}\ \mathrm{N\cdot m}$ on q_3 and $\le 2.57\times 10^{-7}\ \mathrm{N}$ on d_4.

2. Standard 8-Test Verification Matrix (Tests A – G + Check 0)

Test ID	Primary Input Archive	External Load Profile \tau_L(t) = [\tau_{L,3}(t);\ F_{L,4}(t)]	Evaluation Window & Sample Count	Primary Acceptance Metrics (v2.0.0-PRELIMINARY Verified Values)
Check 0 (Raw_SLX)	M2_100kN_0p1111Hz_sp1p50.mat	Direct Simscape 100 kN bail load (P.model_has_coulomb = false)	t \in [5.0, 30.0]\ \mathrm{s} (N=751)	Branch A & B identical (\max\vert{}\hat{\tau}_{L,A} - \hat{\tau}_{L,B}\vert{} = 0); \mathrm{RMSE}_3 = 16{,}694.23\ \mathrm{N\cdot m} (2.27\% of peak), \mathrm{RMSE}_4 = 1{,}766.46\ \mathrm{N} (2.04\% of peak)
Test A (Zero_Load)	M2_null.mat	\tau_L(t) \equiv [0;\ 0] with regularized F_c(\dot{q}) active in plant	t \in [5.0, 30.0]\ \mathrm{s} (N=751)	Branch A \mathrm{RMSE}_3 = 53.47\ \mathrm{N\cdot m}, \mathrm{RMSE}_4 = 4.36\ \mathrm{N}; Branch B \mathrm{RMSE}_3 = 300.43\ \mathrm{N\cdot m} (\le 340), \mathrm{RMSE}_4 = 472.84\ \mathrm{N} (\le 480); 100% bound coverage
Test B1 (Const_Rev)	M2_null.mat	Constant \tau_L = [+220\ \mathrm{kN\cdot m};\ +42\ \mathrm{kN}] across 3 reversals (t = 5.1, 15.1, 25.1\ \mathrm{s})	t \in [5.0, 30.0]\ \mathrm{s} (N=751)	Branch A \mathrm{MAE}_3 = 38.68\ \mathrm{N\cdot m}, \mathrm{MAE}_4 = 3.21\ \mathrm{N}; Branch B

Test ID	Primary Input Archive	External Load Profile \tau_L(t) = [\tau_{L,3}(t);\ F_{L,4}(t)]	Evaluation Window & Sample Count	Primary Acceptance Metrics (v2.0.0-PRELIMINARY Verified Values)
				\mathrm{MAE}_3 = 297.62\ \mathrm{N\cdot m}, \mathrm{MAE}_4 = 471.62\ \mathrm{N}; CovFricPct = 100.0%
Test B2 (Const_Uni)	M2_null.mat	Constant \tau_L = [+220\ \mathrm{kN\cdot m};\ +42\ \mathrm{kN}] on strictly unidirectional segment (\dot{q}_3 > 0, \dot{d}_4 > 0)	t \in [15.5, 24.5]\ \mathrm{s} (N=271, 0 reversals)	Isolates unattenuated Branch B steady-state bias: \mathrm{Bias}_3 = -304.10\ \mathrm{N\cdot m} (-340.00 + 35.90), \mathrm{Bias}_4 = -475.69\ \mathrm{N} (-480.00 + 4.31) vs. Branch A +35.90\ \mathrm{N\cdot m}, +4.31\ \mathrm{N}
Test C (Sinusoidal)	M2_100kN_0p1111Hz_sp1p50.mat	Actual 100 kN 1/9\ \mathrm{Hz} bail load (Q_{q_3,\mathrm{peak}} = 734.24\ \mathrm{kN\cdot m}, Q_{d_4,\mathrm{peak}} = 86.52\ \mathrm{kN})	t \in [5.0, 30.0]\ \mathrm{s} (N=751)	Phase lag 50.05\ \mathrm{ms} (2.002^\circ \approx \arctan(\omega_F/20)), amplitude error -0.065\% (q_3) and -0.070\% (d_4); CovFricPct = 100.0%, CovLagPct = 98.14% (q_3) / 100.0% (d_4), CovRawPct =

Test ID	Primary Input Archive	External Load Profile \tau_L(t) = [\tau_{L,3}(t);\ F_{L,4}(t)]	Evaluation Window & Sample Count	Primary Acceptance Metrics (v2.0.0-PRELIMINARY Verified Values)
				1.73% (q_3) / 18.38% (d_4)
Test D (Step_Load)	M2_null.mat	Step load \Delta\tau_L = [+260\ \mathrm{kN\cdot m};\ +52\ \mathrm{kN}] applied at t_{\mathrm{step}} = 8.50\ \mathrm{s}	t \in [5.0, 30.0]\ \mathrm{s} (N=751)	Theoretical 95\% rise time -\ln(0.05)/20 = 0.1498\ \mathrm{s} (4.49 samples); realized discrete 95\% convergence at 5 samples (0.1667\ \mathrm{s}) on both channels
Test E (Gain_Sweep)	M2_100kN_0p1111Hz_sp1p50.mat	Sweep k_3 = k_4 \in \{5, 10, 20, 40\}\ \mathrm{s}^{-1} on 100 kN bail load	t \in [5.0, 30.0]\ \mathrm{s} (N=751)	Quantifies tracking lag (198.9\ \mathrm{ms} \to 25.0\ \mathrm{ms}) vs. velocity noise amplification \Vert{}K_O D(q)\sigma_{\dot{q}}\Vert{}
Test F (Substep)	M2_100kN_0p1111Hz_sp1p50.mat	Sweep M \in \{1, 2, 5, 10, 20, 50, 67, 100, 200\} for RK4 and Forward Euler	t \in [5.0, 30.0]\ \mathrm{s} (N=751)	RK4 at M=1 is within 0.91\ \mathrm{N\cdot m} of M=200 reference and at M=67 is within 1.96\times 10^{-4}\ \mathrm{N\cdot m}; Forward Euler at M=1 has 10{,}699.4\ \mathrm{N\cdot m} integration error
Test G (Noise_Stress)	M2_100kN_0p1111Hz_sp1p50.mat	Uncalibrated Synthetic Gaussian Noise Stress Test (seed = 202603,	t \in [5.0, 30.0]\ \mathrm{s} (N=751)	Clearly labeled synthetic stress test; demonstrates inertia amplification k_3

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		\sigma_{\dot{q}_3} = 3\ \mathrm{mrad/s}, \sigma_{\dot{d}_4} = 3\ \mathrm{mm/s})		D_{11}(d_4)\sigma_{\dot{q}_3} \approx 180.05\ \mathrm{kN\cdot m} on q_3 vs. 4.74\ \mathrm{kN} on d_4 at k=20\ \mathrm{s}^{-1}

3. Promotion Criteria from PRELIMINARY to FINAL

All M3 outputs remain stamped **PRELIMINARY** until:

1. m2_p05_config_sweep.m ('RunSimulinkGrid' mode) and m2_p08_held_out_validation.m ('RunTorqueDrivenSimulink' mode) are executed in local MATLAB/Simulink R2025b with a clean git status (gitDirty = "clean").
2. m3_observer_suite.m is run against the newly generated MS02P05 .mat archives and passes all 7 ingestion checks and Test C/E/F bounds.